Vidyavardhini's College of Engineering and Technology, Vasai
Department of Computer Science & Engineering (Data Science)

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Subject: MDM- Wireless & Mobile Communication

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Title:	To analyze channel interference in the 2.4 GHz Wi-Fi band by identifying nearby Wi-Fi networks, recording their operating channels and RSSI values, and evaluating channel occupancy and overlap.
Date of Performance:	
Date of Submission:	
Marks:	
Sign of Faculty:	

Aim: To analyze channel interference in the 2.4 GHz Wi-Fi band by identifying nearby Wi-Fi networks, recording their operating channels and RSSI values, and evaluating channel occupancy and overlap.

Objective: The objectives of this experiment are:

- To identify nearby Wi-Fi networks operating in the 2.4 GHz frequency band.
- To identify the SSID and operating channel of detected Wi-Fi networks.
- To measure and record the RSSI (Received Signal Strength Indicator) of nearby networks.
- To observe the distribution of Wi-Fi networks across different 2.4 GHz channels.
- To identify crowded and less-crowded channels based on channel occupancy.
- To analyse the effect of adjacent-channel overlap on Wi-Fi interference.
- To compare the signal strengths of networks operating on the same or overlapping channels.
- To identify channels that are likely to experience higher interference.

Theory:

1. **2.4 GHz Wi-Fi Band**

Wi-Fi networks can operate in different frequency bands. The 2.4 GHz band is widely used because it provides relatively good coverage and penetration through obstacles.
In the 2.4 GHz band, Wi-Fi networks operate on multiple channels. Since these channels are closely spaced, neighbouring channels can partially overlap with one another.

2. **Wi-Fi Channel**

A Wi-Fi channel represents a specific portion of the radio-frequency spectrum used by a wireless network for communication.
For this experiment, the commonly observed 2.4 GHz channels are:

Channel 1, Channel 6 and Channel 11

These channels are commonly preferred because they have minimal overlap with one another when using standard 20 MHz channel widths.

3. Channel Interference

Channel interference occurs when multiple Wi-Fi networks operate on the same or overlapping frequency channels.

For example:

Channel 1 Channel 6 Channel 11

↓ ↓ ↓

Wi-Fi A Wi-Fi B Wi-Fi C

If several nearby networks use the same channel, they compete for the available wireless medium. This can result in:

- Reduced network performance
- Increased packet retransmissions
- Lower throughput
- Higher delay
- Unstable wireless connectivity

4. Adjacent-Channel Interference

Adjacent-channel interference occurs when Wi-Fi networks use channels whose frequency ranges overlap.

For example:

Channel 1

Channel 3

Channel 6

The overlapping portions can contribute to interference.

5. RSSI

RSSI (Received Signal Strength Indicator) indicates the received strength of a Wi-Fi signal and is generally represented in dBm.

RSSI values are normally negative.

Remember:

- Closer to 0 dBm → Stronger signal
- More negative value → Weaker signal

For example:

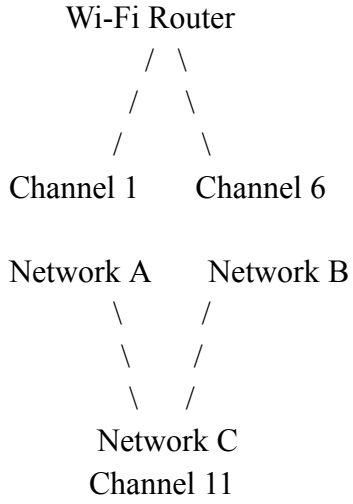
-40 dBm → Strong signal

-55 dBm → Moderate signal
-70 dBm → Weak signal
-85 dBm → Very weak signal
A strong nearby network operating on the same or overlapping channel is more likely to contribute significantly to interference.

4. Experimental Setup
Required Equipment / Software

- Smartphone
- Wi-Fi Analyzer application
- Nearby 2.4 GHz Wi-Fi networks
- Wi-Fi router/access point
- Laboratory notebook for recording observations

Experimental Arrangement



The smartphone is used to scan nearby wireless networks and observe their **SSID, channel, frequency band and RSSI**.

5. Step-by-Step Procedure

Step 1: Install Wi-Fi Analyzer
Install and open a suitable **Wi-Fi Analyzer application** on the smartphone.

Step 2: Enable Wi-Fi
On the smartphone:

1. Open Settings.

2. Select Wi-Fi.
3. Turn Wi-Fi ON.
4. Turn Location ON if required by the analyzer application.
5. Provide the required permissions.

Step 3: Start Wi-Fi Scan

Open the Wi-Fi Analyzer application and start scanning for nearby wireless networks.
The application may display information such as:
SSID : MyWiFi
Frequency : 2.4 GHz
Channel : 6
RSSI : -48 dBm

Step 4: Select the 2.4 GHz Band

Select or filter the scan results to display networks operating in the 2.4 GHz band.

Step 5: Record SSID

Observe the list of detected Wi-Fi networks and record the SSID (network name) of each network.

Step 6: Record Channel Number

Record the channel number used by each detected network.
For example:
Network A → Channel 1
Network B → Channel 6
Network C → Channel 11

Step 7: Record RSSI

Record the RSSI value of every detected network in dBm.
For example:
Network A → -45 dBm
Network B → -62 dBm
Network C → -74 dBm

Step 8: Open Channel Graph

Open the Channel Graph, Channel Rating, or equivalent visualization provided by the Wi-Fi Analyzer.
Observe:

- Number of networks on each channel
- Signal strength of individual networks

- Overlap between neighboring channels
- Strong and weak signals

Step 9: Identify Crowded Channels
Count the number of networks operating on each channel.
The channel having a larger number of nearby networks can be considered more crowded.
Step 10: Identify Channel Overlap
Observe the channel graph and identify networks whose signal regions overlap.
Pay particular attention to networks operating on the same or nearby channels.

Step 11: Compare RSSI
Compare the RSSI values of networks operating on the same or overlapping channels.
A network with a stronger RSSI is likely to have a greater impact on the local wireless environment than a very weak network.

Step 12: Analyse Interference
Identify channels likely to experience higher interference based on:

- Number of networks
- Channel occupancy
- Channel overlap
- RSSI/signal strength

Step 13: Record Observations
Enter all observations in the tables provided below.

Step 14: Write Conclusion
Based on the observations, identify the **most crowded channel, least crowded channel, and channels likely to experience higher interference**.

6. Observation Table

Sr. No.	SSID	RSSI (dBm)	Channel No.	Frequency Band	Interference Observation
1	GSVCET	-45	11	2.4 GHz	low interference
2	Exam Section	-57	1	2.4 GHz	Good Signal
3	OM 13R	-61	11	2.4 GHz	Moderate signal
4	Oppo F25	-80	4	2.4 GHz	Very Weak

Sr. No.	SSID	RSSI (dBm)	Channel No.	Frequency Band	Interference Observation
5	RD	-75	1	2.4 GHz	Weak Signal
6	Dinesh	-80	1	2.4 GHz	Very Weak

Note: Add additional rows if more than five networks are detected.

7. Channel Analysis

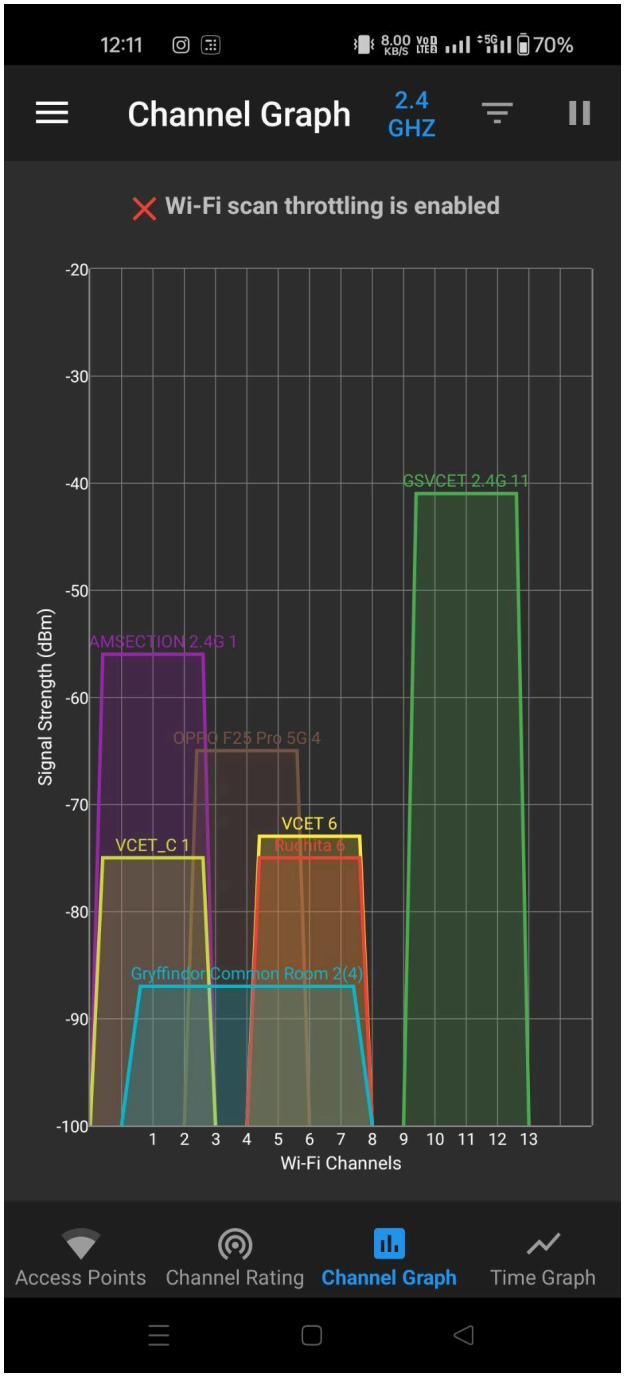
Channel No.	No. of Networks	Strongest RSSI (dBm)	Channel Occupancy	Interference Level
1	3	-57	Medium	Medium
4	1	-80	Low	Low
11	2	-45	Medium	High

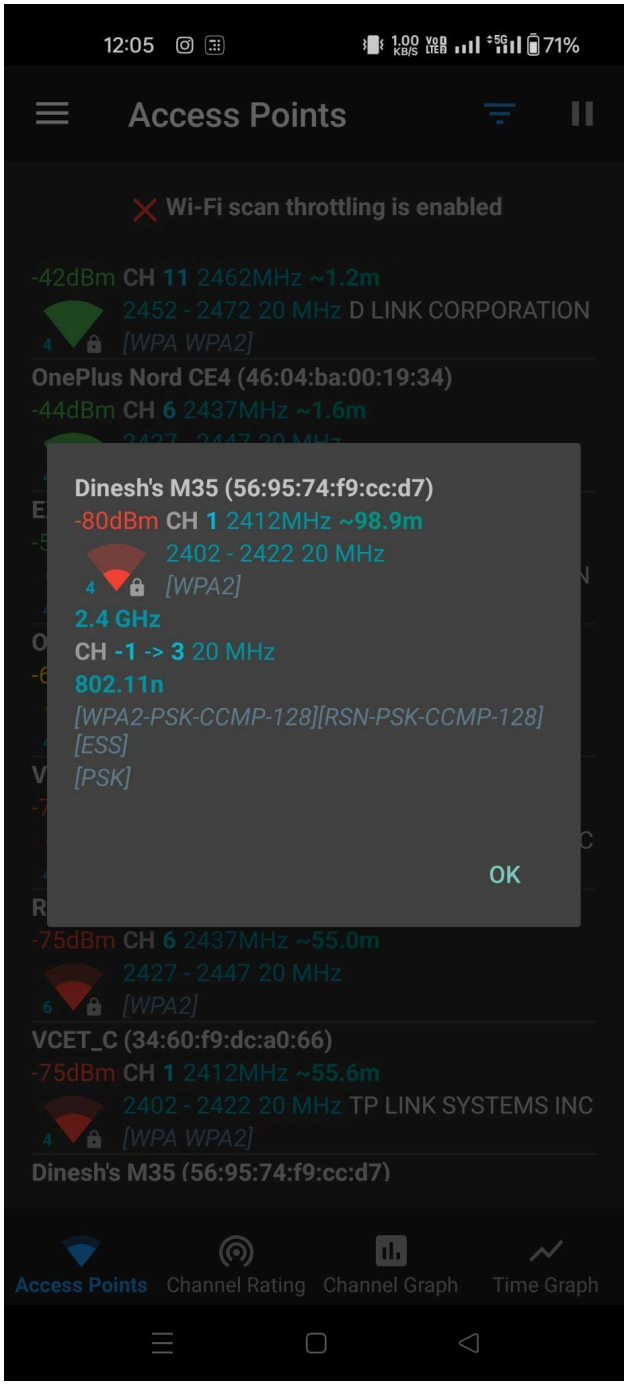
8. Channel Interference Analysis

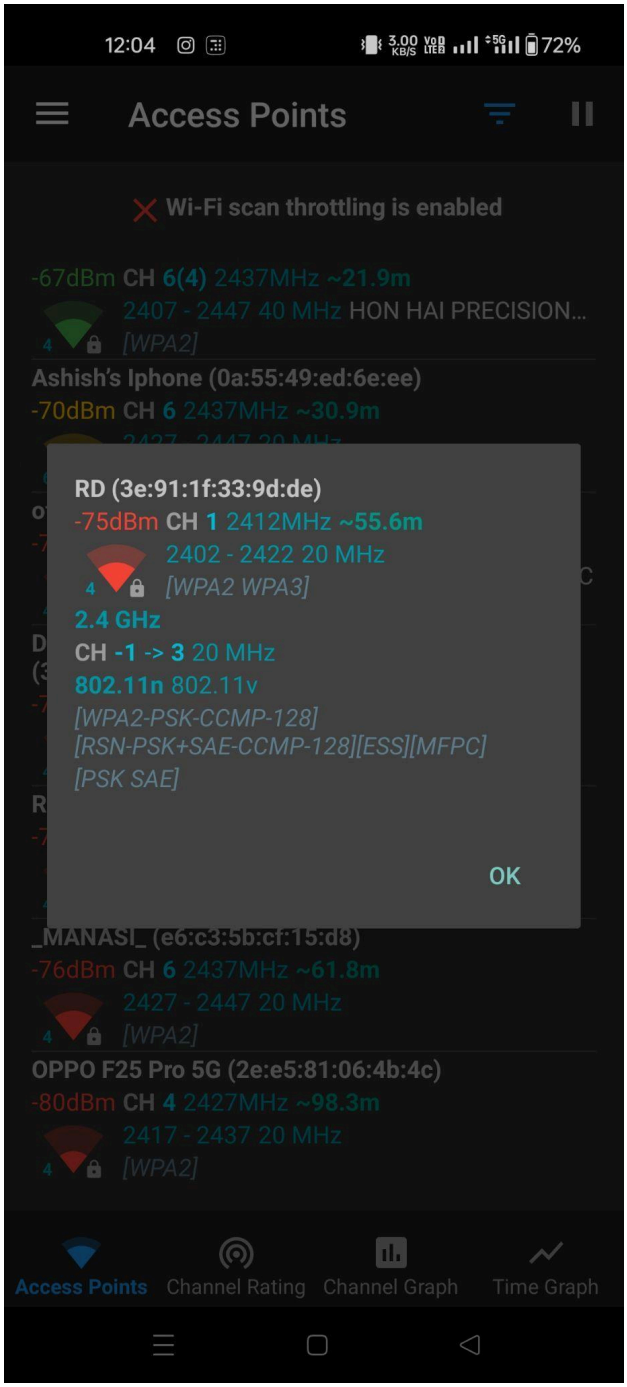
Record the following observations after examining the Channel Graph.

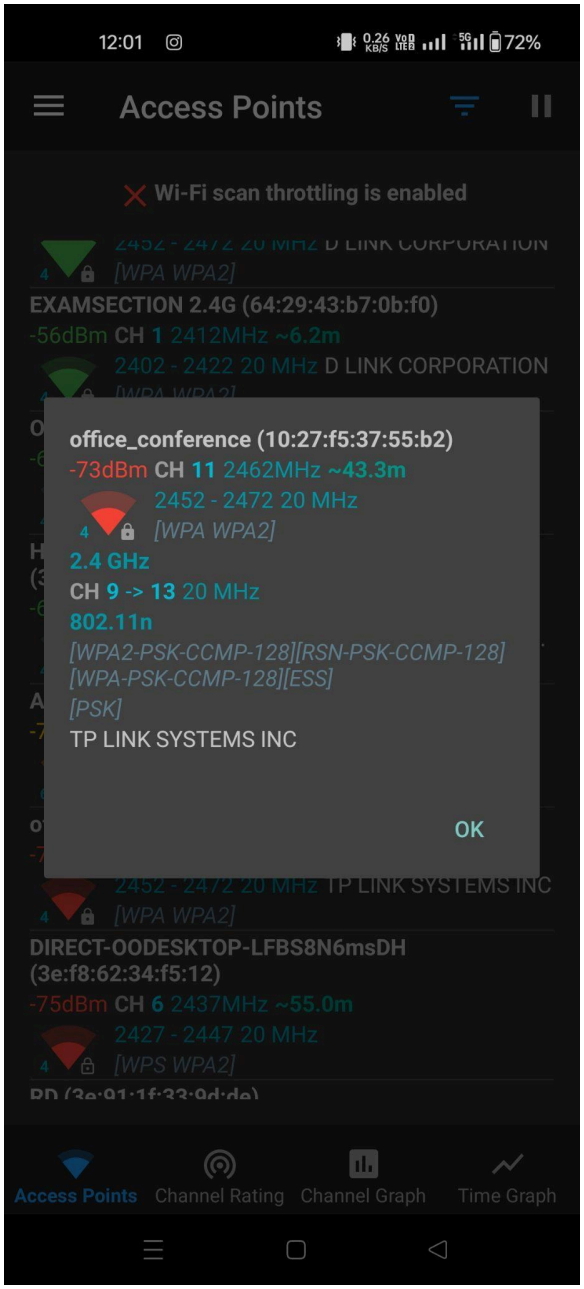
Parameter	Observation
Most crowded channel	channel 2/3
Least crowded channel	channel 9-13
Channel with strongest nearby network	channel 11
Channel with weakest nearby network	channel 1
Channel showing maximum overlap	channel 1-8
Channel likely to experience highest interference	channel 1-6
Channel likely to experience lowest interference	channel 9-13

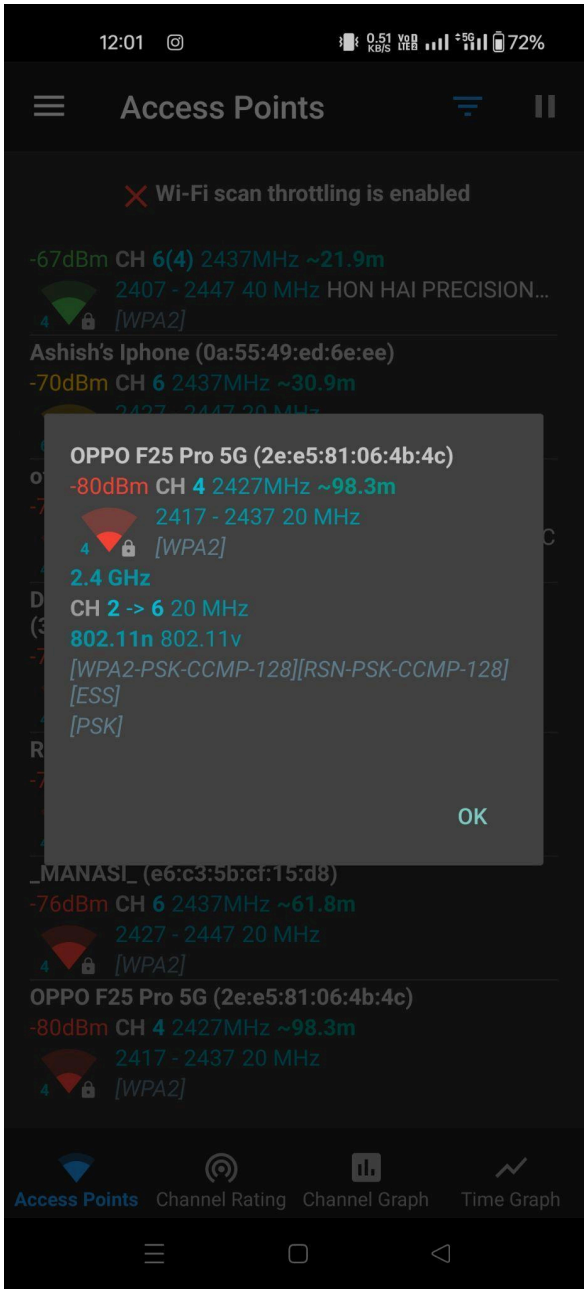
Code/Output:

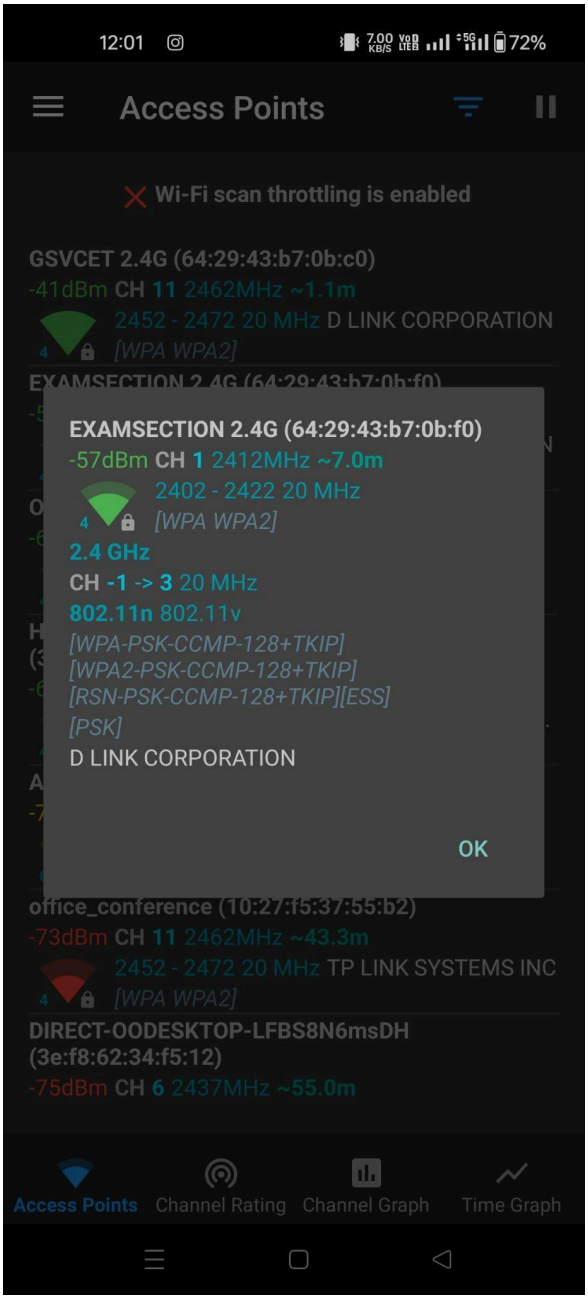

12:11
70%
Channel Graph
2.4 GHZ
Wi-Fi scan throttling is enabled
-20
-30
-40
-50
-60
-70
-80
-90
-100
Signal Strength (dBm)
GSVCET 2.4G 11
AMSECTION 2.4G 1
OPPO F25 Pro 5G 4
VCET 6
Ruchita 6
VCET_C 1
Gryffindor Common Room 2(4)
1 2 3 4 5 6 7 8 9 10 11 12 13
Wi-Fi Channels
Access Points
Channel Rating
Channel Graph
Time Graph

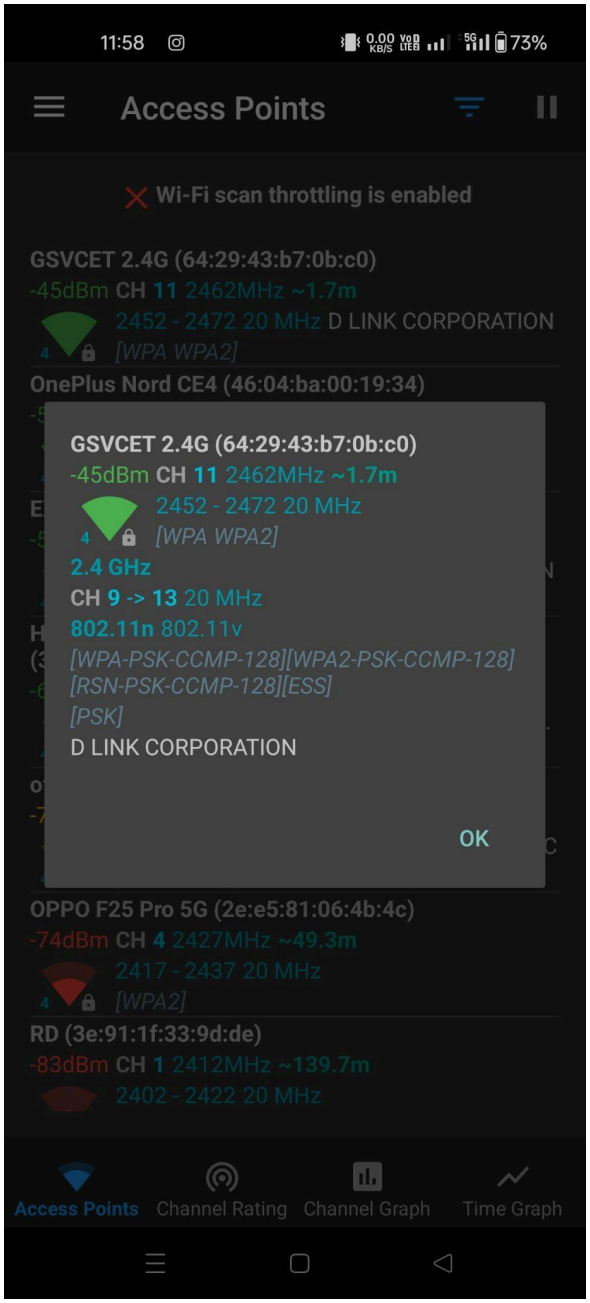

12:05
71%
Access Points
Wi-Fi scan throttling is enabled
-42dBm CH 11 2462MHz ~1.2m
2452 - 2472 20 MHz D LINK CORPORATION
[WPA WPA2]
OnePlus Nord CE4 (46:04:ba:00:19:34)
-44dBm CH 6 2437MHz ~1.6m
Dinesh's M35 (56:95:74:f9:cc:d7)
-80dBm CH 1 2412MHz ~98.9m
2402 - 2422 20 MHz
[WPA2]
2.4 GHz
CH -1 -> 3 20 MHz
802.11n
[WPA2-PSK-CCMP-128][RSN-PSK-CCMP-128]
[ESS]
[PSK]
OK
-75dBm CH 6 2437MHz ~55.0m
2427 - 2447 20 MHz
[WPA2]
VCET_C (34:60:f9:dc:a0:66)
-75dBm CH 1 2412MHz ~55.6m
2402 - 2422 20 MHz TP LINK SYSTEMS INC
[WPA WPA2]
Dinesh's M35 (56:95:74:f9:cc:d7)
Access Points
Channel Rating
Channel Graph
Time Graph


12:04
72%
Access Points
Wi-Fi scan throttling is enabled
-67dBm CH 6(4) 2437MHz ~21.9m
2407 - 2447 40 MHz HON HAI PRECISION...
[WPA2]
Ashish's Iphone (0a:55:49:ed:6e:ee)
-70dBm CH 6 2437MHz ~30.9m
RD (3e:91:1f:33:9d:de)
-75dBm CH 1 2412MHz ~55.6m
2402 - 2422 20 MHz
[WPA2 WPA3]
2.4 GHz
CH -1 -> 3 20 MHz
802.11n 802.11v
[WPA2-PSK-CCMP-128]
[RSN-PSK+SAE-CCMP-128][ESS][MFPC]
[PSK SAE]
OK
MANASI (e6:c3:5b:cf:15:d8)
-76dBm CH 6 2437MHz ~61.8m
2427 - 2447 20 MHz
[WPA2]
OPPO F25 Pro 5G (2e:e5:81:06:4b:4c)
-80dBm CH 4 2427MHz ~98.3m
2417 - 2437 20 MHz
[WPA2]
Access Points
Channel Rating
Channel Graph
Time Graph


12:01
72%
Access Points
Wi-Fi scan throttling is enabled
[WPA WPA2]
EXAMSECTION 2.4G (64:29:43:b7:0b:f0)
-56dBm CH 1 2412MHz ~6.2m
2402 - 2422 20 MHz D LINK CORPORATION
office_conference (10:27:f5:37:55:b2)
-73dBm CH 11 2462MHz ~43.3m
2452 - 2472 20 MHz
[WPA WPA2]
2.4 GHz
CH 9 -> 13 20 MHz
802.11n
[WPA2-PSK-CCMP-128][RSN-PSK-CCMP-128]
[WPA-PSK-CCMP-128][ESS]
[PSK]
TP LINK SYSTEMS INC
OK
DIRECT-OODESKTOP-LFBS8N6msDH
(3e:f8:62:34:f5:12)
-75dBm CH 6 2437MHz ~55.0m
2427 - 2447 20 MHz
[WPS WPA2]
Access Points
Channel Rating
Channel Graph
Time Graph


12:01
72%
Access Points
Wi-Fi scan throttling is enabled
-67dBm CH 6(4) 2437MHz ~21.9m
2407 - 2447 40 MHz HON HAI PRECISION...
[WPA2]
Ashish's Iphone (0a:55:49:ed:6e:ee)
-70dBm CH 6 2437MHz ~30.9m
OPPO F25 Pro 5G (2e:e5:81:06:4b:4c)
-80dBm CH 4 2427MHz ~98.3m
2417 - 2437 20 MHz
[WPA2]
2.4 GHz
CH 2 -> 6 20 MHz
802.11n 802.11v
[WPA2-PSK-CCMP-128][RSN-PSK-CCMP-128]
[ESS]
[PSK]
OK
MANASI (e6:c3:5b:cf:15:d8)
-76dBm CH 6 2437MHz ~61.8m
2427 - 2447 20 MHz
[WPA2]
OPPO F25 Pro 5G (2e:e5:81:06:4b:4c)
-80dBm CH 4 2427MHz ~98.3m
2417 - 2437 20 MHz
[WPA2]
Access Points
Channel Rating
Channel Graph
Time Graph


12:01
72%
Access Points
Wi-Fi scan throttling is enabled
GSVCET 2.4G (64:29:43:b7:0b:c0)
-41dBm CH 11 2462MHz ~1.1m
2452 - 2472 20 MHz D LINK CORPORATION
[WPA WPA2]
EXAMSECTION 2.4G (64:29:43:b7:0b:f0)
-57dBm CH 1 2412MHz ~7.0m
2402 - 2422 20 MHz
[WPA WPA2]
2.4 GHz
CH -1 -> 3 20 MHz
802.11n 802.11v
[WPA-PSK-CCMP-128+TKIP]
[WPA2-PSK-CCMP-128+TKIP]
[RSN-PSK-CCMP-128+TKIP][ESS]
[PSK]
D LINK CORPORATION
OK
office_conference (10:27:f5:37:55:b2)
-73dBm CH 11 2462MHz ~43.3m
2452 - 2472 20 MHz TP LINK SYSTEMS INC
[WPA WPA2]
DIRECT-OODESKTOP-LFBS8N6msDH
(3e:f8:62:34:f5:12)
-75dBm CH 6 2437MHz ~55.0m
Access Points
Channel Rating
Channel Graph
Time Graph



Conclusion: Channel analysis indicates that the 2.4 GHz band is congested.